

O4_Engineering_Standards.md

Part 2 — Frontend Engineering Standards

Frontend Architecture

The frontend follows:

- Component-Driven Development*
 - Feature-Based Architecture*
 - Server-First Rendering*
 - Type-Safe Development*
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React Standards

Rules

- Functional Components only*
 - No Class Components*
 - One component per file*
 - Prefer composition over inheritance*
 - Keep components small and reusable*
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Next.js Standards

Use

- App Router
 - Server Components by default
 - Client Components only when required
 - Route Groups for organization
 - Nested Layouts
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TypeScript Standards

Rules

- Strict Mode enabled
 - No any
 - Prefer interfaces for objects
 - Use enums only when necessary
 - Explicit return types for exported functions
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Component Architecture

Structure

text id="5hhg7r" Feature | | Components | Hooks | Services |
Types | Utils | Constants

Component Rules

Every component must:

- Have a single responsibility
 - Be reusable
 - Accept typed props
 - Avoid unnecessary state
 - Be responsive
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Props Standards

Rules

- Typed interfaces
 - Required props first
 - Optional props last
 - Avoid prop drilling
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State Management

Priority

1. Local State (useState)
2. Context API
3. Global Store (Zustand)

Rules

- Keep state as local as possible

- *Avoid duplicate state*
 - *Derived state should not be stored*
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Data Fetching

Use

- *Server Components*
- *Server Actions*
- *Fetch API*

Rules

- *Avoid client-side fetching unless necessary*
 - *Cache static data*
 - *Revalidate dynamic data when needed*
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Forms

Preferred Libraries

- *React Hook Form*
- *Zod*

Rules

- *Validate on client*
 - *Validate again on server*
 - *Display clear validation messages*
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Styling Standards

Use

- *Tailwind CSS*
- *Design Tokens*
- *CSS Variables*

Rules

- *No inline styles*
 - *No hardcoded colors*
 - *No duplicated utility classes*
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Animation Standards

Library

- *Framer Motion*

Rules

- *60 FPS target*
 - *Meaningful animations only*
 - *Respect reduced motion preferences*
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Error Handling

Every page should handle:

- *Loading*

- *Empty State*
 - *Error State*
 - *Success State*
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Accessibility Standards

Requirements

- *Semantic HTML*
 - *Keyboard Navigation*
 - *ARIA Labels*
 - *Focus Management*
 - *Accessible Color Contrast*
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Performance Standards

Targets

- *Lazy load heavy components*
 - *Optimize images*
 - *Dynamic import where appropriate*
 - *Minimize client-side JavaScript*
 - *Avoid unnecessary re-renders*
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Code Quality

Every feature must:

- Pass TypeScript checks
 - Pass ESLint
 - Pass Prettier
 - Build successfully
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Frontend Checklist

Before Merge

- Responsive
 - Accessible
 - Type Safe
 - Performance Optimized
 - Design System Compliant
 - No Console Errors
 - No Type Errors
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Related Documents

- 09_Design_System.md
 - 10_Database_Architecture.md
 - 11_API_Architecture.md
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Approval Status

Version: 1.0.0

Status: Draft

Next Document

04_Engineering_Standards.md — Part 3